# SOFTWARE ENGINEERING LAB 07


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## SECTION A CSE SEM 5

## SRN: PES1UG24CS020


## 1 . Running the Buggy Code:

```
(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD
Traceback (most recent call last):
  File "C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7\gym_membership.py", line 30, in <module>
    main()
  File "C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7\gym_membership.py", line 16, in main
    result = parse_plan(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^
  File "C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7\gym_membership.py", line 2, in parse_plan
    plan, months = code.split('-')
    ^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>
```

## 2 . Running test_gym_membership.py

```
test_gym_membership.py:104:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_gym_membership.py:105: in test_average_workout_empty_rejected
    assert average_workout(s) is None
           ^^^^^^^^^^^^^^^^^^
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

minutes = ''

    def average_workout(minutes):
        values = [int(m) for m in minutes.split()]
>       return sum(values) // len(values)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^
E       ZeroDivisionError: integer division or modulo by zero
E       Failing test case: test_average_workout_empty_rejected(
E           s='',  # or any other generated value
E       )

gym_membership.py:11: ZeroDivisionError
=========================================================== short test summary info ============================================================
FAILED test_gym_membership.py::test_parse_plan_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_calculate_bmi_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_average_workout_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_parse_plan_any_input_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_calculate_bmi_any_input_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_average_workout_any_input_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_parse_plan_missing_dash_rejected - ValueError: not enough values to unpack (expected 2, got 1)
FAILED test_gym_membership.py::test_calculate_bmi_zero_height_rejected - ZeroDivisionError: float division by zero
FAILED test_gym_membership.py::test_average_workout_empty_rejected - ZeroDivisionError: integer division or modulo by zero
============================================================ 9 failed, 3 passed in 11.08s =============================================================

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>
```

## 3 . Fixing Buggy Application Python Script

```
(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD
Invalid plan code. Expected format: NAME-MONTHS (e.g. GOLD-12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,0
Invalid input. Expected: weight,height as positive numbers (e.g. 70,1.75)

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30):
Invalid input. Expected: non-negative whole numbers separated by spaces

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>
```

# 4 . Rerun test_gym_membership.py

```
(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python -m pytest test_gym_membership.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7
plugins: hypothesis-6.170.14
collected 12 items

test_gym_membership.py ............                                                                                                    [100%]

============================================================ 12 passed in 3.98s =============================================================

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>python -m pytest test_gym_membership.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0 -- C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7
plugins: hypothesis-6.170.14
collected 12 items

test_gym_membership.py::test_parse_plan_no_crash PASSED                                                                               [  8%]
test_gym_membership.py::test_calculate_bmi_no_crash PASSED                                                                            [ 16%]
test_gym_membership.py::test_average_workout_no_crash PASSED                                                                          [ 25%]
test_gym_membership.py::test_parse_plan_any_input_no_crash PASSED                                                                     [ 33%]
test_gym_membership.py::test_calculate_bmi_any_input_no_crash PASSED                                                                  [ 41%]
test_gym_membership.py::test_average_workout_any_input_no_crash PASSED                                                                [ 50%]
test_gym_membership.py::test_parse_plan_valid PASSED                                                                                  [ 58%]
test_gym_membership.py::test_calculate_bmi_valid PASSED                                                                               [ 66%]
test_gym_membership.py::test_average_workout_valid PASSED                                                                             [ 75%]
test_gym_membership.py::test_parse_plan_missing_dash_rejected PASSED                                                                  [ 83%]
test_gym_membership.py::test_calculate_bmi_zero_height_rejected PASSED                                                                [ 91%]
test_gym_membership.py::test_average_workout_empty_rejected PASSED                                                                    [100%]

============================================================ 12 passed in 2.16s =============================================================

(.venv) C:\3rd year material\SE\SE_SUBMISSION_PES1UG24CS020\LAB7>
```